

Alaoglu–Erdős Beyond Report A

S-unit secant heights, primitive residual cancellation, fixed-order extrapolation, and reciprocal cyclotomic barriers

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Abstract

The Alaoglu–Erdős integer-exponent conjecture asks whether

$$2^x, 3^x \in \mathbb{Z} \implies x \in \mathbb{Z}$$

for real x . The attached Report A already develops the exact conditional monoid, finite verification, determinant hierarchies, p -adic assignment methods, rational and algebraic output spectra, interpolation, operator, Mahler, moment, and canonical-product approaches. This continuation deliberately does not repeat those lines.

Assuming a least nonintegral solution β , with $M = 2^\beta$ and $A = 3^\beta$, we normalize continued-fraction convergents $r_j/s_j \rightarrow \beta$ into rational S -unit points

$$(u_j, v_j) = \left(\frac{2^{r_j}}{M^{s_j}}, \frac{3^{r_j}}{A^{s_j}} \right) = (2^{\varepsilon_j}, 3^{\varepsilon_j}) \longrightarrow (1, 1), \quad \varepsilon_j = r_j - s_j\beta.$$

Corvaja–Zannier height theory then gives an exact linear asymptotic for the reduced height of the rational secant $(u_j - 1)/(v_j - 1)$. The coefficient is an explicit valuation functional $\kappa(M, A; \beta) > 0$. Consequently the reduced numerator and denominator of the secant are each $\exp((\kappa + o(1))s_j)$, while its error as an approximation to $\log 2 / \log 3$ is only polynomial in s_j . This produces a sharp “height exponent zero” barrier. It also yields an exact asymptotic for the total cross-residual gcd and shows that its entire exponential part is supported on the fixed primes dividing $6MA$; all auxiliary prime cancellation is subexponential.

The same mechanism rules out every fixed-order Richardson/Prouhet extrapolation of toric secants: order m improves the analytic error to $\Theta(|\varepsilon_j|^m)$, but the reduced height remains $\Theta(s_j)$. A separate algebraic argument proves that two distinct monic reciprocal polynomials of the same degree can agree at infinity to at most half their degree. For equal-totient cyclotomic pairs this is an exact universal ceiling. A scan through index 50000 finds 665095 equal-totient pairs, confirms the ceiling, and identifies only 24 sharp pairs in that range.

These results do not prove the conjecture. They do, however, eliminate a broad new class of fixed-complexity secant, finite-difference, common-residual, and reciprocal-cyclotomic amplifiers, and isolate the surviving requirement: a growing-complexity construction whose *primitive* height is saved by output-specific cancellation at the structural primes.

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1 Executive answer and claim ledger

1.1 Outcome

No complete proof is claimed here. The four-exponentials obstruction remains real: a nonintegral solution would make the four algebraic numbers $2, 3, 2^\beta, 3^\beta$ occupy the forbidden 2×2 configuration predicted by the Four Exponentials Conjecture. What is new is a conditional transformation of such a solution into a near-identity rank-two S -unit orbit on which the Subspace Theorem has unusually sharp consequences.

The main conclusions are the following.

1. **Exact secant height.** For convergents r_j/s_j to β , the rational secants

$$\sigma_j = \frac{u_j - 1}{v_j - 1} = \frac{A^{s_j}(2^{r_j} - M^{s_j})}{M^{s_j}(3^{r_j} - A^{s_j})}$$

satisfy

$$h(\sigma_j) = \kappa(M, A; \beta)s_j + o(s_j),$$

where κ is the explicit positive valuation functional in Theorem 3.1. [\[proved here\]](#)

2. **Primitive numerator and denominator.** If $\sigma_j = P_j/Q_j$ in lowest terms, then

$$\log P_j = \log Q_j = \kappa s_j + o(s_j).$$

At the same time

$$\sigma_j - \frac{\log 2}{\log 3} = \Theta(\varepsilon_j), \quad s_j^{-C} \ll |\varepsilon_j| < s_j^{-1}.$$

Thus the approximation exponent relative to reduced height tends to zero. [\[proved here\]](#)

3. **Exact cross-residual cancellation.** With

$$X_j = A^{s_j}|2^{r_j} - M^{s_j}|, \quad Y_j = M^{s_j}|3^{r_j} - A^{s_j}|, \quad G_j = \gcd(X_j, Y_j),$$

one has

$$\log G_j = (\log M + \log A - \kappa)s_j + o(s_j).$$

After removing the fixed primes dividing $6MA$, the remaining common divisor is $\exp(o(s_j))$. Hence every linear-rate cancellation in G_j is structural, not supplied by fresh primes. [\[proved here\]](#)

4. **Fixed-order extrapolation barrier.** Put

$$\phi(t) = \frac{2^t - 1}{3^t - 1}, \quad \phi(0) = \frac{\log 2}{\log 3},$$

and, for fixed $m \geq 1$,

$$R_m(t) = \sum_{i=1}^m (-1)^{i-1} \binom{m}{i} \phi(it).$$

Then $R_m(\varepsilon_j) \in \mathbb{Q}$,

$$R_m(\varepsilon_j) - \phi(0) = \Theta(\varepsilon_j^m), \quad h(R_m(\varepsilon_j)) = \Theta(s_j),$$

and again the height-normalized approximation exponent is zero. This covers every fixed finite-difference order, not merely the first secant. [\[proved here\]](#)

5. **Reciprocal half-contact ceiling.** If $P, Q \in \mathbb{Z}[Z]$ are distinct monic reciprocal polynomials of common degree D and

$$\frac{P(Z)}{Q(Z)} = 1 + cZ^{-K} + O(Z^{-K-1}), \quad c \neq 0,$$

then $K \leq \lfloor D/2 \rfloor$. Equality forces $P - Q$ to be a single middle monomial. For $P = \Phi_n, Q = \Phi_m$ with $\varphi(n) = \varphi(m)$ this bounds the first unequal Möbius layer by $\varphi(n)/2$. [\[proved here\]](#)

6. **Finite cyclotomic reconnaissance.** Among all 665095 pairs $3 \leq n < m \leq 50000$ with $\varphi(n) = \varphi(m)$, the maximum contact ratio is $1/2$, attained by 24 pairs. Every sharp pair found belongs to one of two infinite elementary families or is the exceptional pair $(3, 4)$. No global classification is claimed. [\[computation\]](#)

Central message. The near-identity orbit supplies arbitrarily accurate analytic contact, but fixed rational observables retain linear primitive height. The Subspace Theorem does not merely fail to finish the proof: it identifies exactly where the lost size goes. Any successful continuation must use a family whose complexity grows and whose reduced height enjoys cancellation not available to any fixed rational function on $\mathbb{G}_m^2$.

1.2 Evidence labels

- [\[proved here\]](#) A complete conventional proof is included in this report. No Lean kernel claim is made.
- [\[classical input\]](#) A named theorem from the literature is used, chiefly the Corvaja–Zannier S -unit height theorem and a standard Baker lower bound for a linear form in two logarithms.
- [\[computation\]](#) A finite Python computation is reproduced in Appendix D; it is not a formal proof of an infinite statement.
- [\[open target\]](#) A precise sufficient target or an unresolved extension.

1.3 Public-source reconnaissance

The model-comparison background was checked only to calibrate the competitive context, not to import mathematical assertions. Anthropic’s public launch page for Claude Fable 5 is <https://www.anthropic.com/news/claude-fable-5-mythos-5>. Its 28 July 2026 research post, *Discovering cryptographic weaknesses with Claude*, states that Fable 5 had resolved the Jacobian Conjecture earlier in July; see <https://www.anthropic.com/research/discovering-cryptographic-weaknesses>. I found no public primary document disclosing the claimed Alaoglu–Erdős idea. The mathematics below is therefore independent of any undisclosed rival construction.

1.4 Nonduplication boundary

Report A already treats the exact solution monoid, candidate counting, finite scans, Four/Six Exponentials reductions, prime-log-product independence, arbitrary-node and mixed semigroup

determinants, p -adic assignment cascades, Loewner and matrix-coefficient certificates, rational contact rigidity, geometric Newton interpolation, Mahler and natural boundary diagnostics, operator formulations, moments, Padé defects, carry languages, canonical products, and several denominator barriers. This continuation uses only the minimal conditional normalization from that corpus. Its new object is the convergent-normalized S -unit orbit and the primitive height of rational observables on that orbit.

2 Conditional setup and the near-identity orbit

2.1 The least generator

Assume the conjecture is false. Let $\beta > 0$ be its least nonintegral solution and put

$$M = 2^\beta \in \mathbb{N}, \quad A = 3^\beta \in \mathbb{N}. \quad (1)$$

The inherited exact classification gives all solutions as $n + k\beta$ with $n, k \in \mathbb{N}_0$. Only three elementary consequences are needed here:

1. β is irrational, indeed transcendental;
2. M is not a pure power of 2, and A is not a pure power of 3;
3. $\log M = \beta \log 2$ and $\log A = \beta \log 3$.

Let r_j/s_j be the ordinary continued-fraction convergents to β , with $s_j > 0$, and set

$$\varepsilon_j = r_j - s_j\beta. \quad (2)$$

Then $\varepsilon_j \neq 0$, $\varepsilon_j \rightarrow 0$, and

$$|\varepsilon_j| < \frac{1}{s_j}. \quad (3)$$

A lower bound for the nonzero linear form

$$r_j \log 2 - s_j \log M = \varepsilon_j \log 2$$

in two logarithms of algebraic numbers gives constants $C_0, C_1 > 0$, depending only on M , such that

$$|\varepsilon_j| \geq C_0 s_j^{-C_1} \quad (4)$$

for all j . This is the only quantitative transcendence estimate used below.

2.2 The normalized S -unit point

Define

$$u_j = \frac{2^{r_j}}{M^{s_j}}, \quad v_j = \frac{3^{r_j}}{A^{s_j}}. \quad (5)$$

By (1) and (2), as positive real numbers,

$$u_j = 2^{\varepsilon_j}, \quad v_j = 3^{\varepsilon_j}. \quad (6)$$

Thus $(u_j, v_j) \rightarrow (1, 1)$ along the analytic one-parameter subgroup $t \mapsto (2^t, 3^t)$, while each coordinate is a rational number supported on the fixed set of primes dividing $6MA$.

Lemma 2.1 (Multiplicative independence). *For every j , the rational numbers u_j and v_j are multiplicatively independent.*

Proof. Suppose $u_j^a v_j^b = 1$ with $a, b \in \mathbb{Z}$. Taking the real logarithm and using $\varepsilon_j \neq 0$ gives

$$a \log 2 + b \log 3 = 0.$$

Hence $2^a 3^b = 1$, so unique factorization gives $a = b = 0$. □

The points lie in the finitely generated subgroup

$$\Gamma = \langle (2, 3), (M^{-1}, A^{-1}) \rangle \subset \mathbb{G}_m^2(\mathbb{Q}). \quad (7)$$

This is precisely the setting of Corvaja–Zannier’s height theorem.

2.3 Coordinate heights

Write

$$M = 2^m M_0, \quad 2 \nmid M_0, \quad A = 3^a A_0, \quad 3 \nmid A_0. \quad (8)$$

The nonintegrality of β implies $M_0 > 1$ and $A_0 > 1$. Since $r_j/s_j \rightarrow \beta$, we have $r_j > ms_j$ and $r_j > as_j$ for all sufficiently large j . The fractions

$$u_j = \frac{2^{r_j - ms_j}}{M_0^{s_j}}, \quad v_j = \frac{3^{r_j - as_j}}{A_0^{s_j}}$$

are then in lowest terms. Consequently

$$h(u_j) = s_j \log M_0 + O(|\varepsilon_j|), \quad (9)$$

$$h(v_j) = s_j \log A_0 + O(|\varepsilon_j|). \quad (10)$$

In particular both heights tend linearly to infinity even though the real values tend to 1. This archimedean/nonarchimedean split is the source of the method.

3 The exact projective-height slope

For a rational projective point, use the absolute logarithmic Weil height

$$h(1 : u : v) = \sum_w \log \max\{1, |u|_w, |v|_w\},$$

with the usual normalized absolute values on $\mathbb{Q}$.

For every rational prime p , put

$$m_p = v_p(M), \quad a_p = v_p(A),$$

and let $\mathbf{1}_{p=q}$ denote the indicator of $p = q$.

Theorem 3.1 (Exact projective-height slope). *Define*

$$\kappa(M, A; \beta) = \sum_{p \text{ prime}} \max\{0, m_p - \beta \mathbf{1}_{p=2}, a_p - \beta \mathbf{1}_{p=3}\} \log p. \quad (11)$$

Then

$$h(1 : u_j : v_j) = \kappa(M, A; \beta) s_j + o(s_j). \quad (12)$$

Moreover

$$\kappa \geq \max\{\log M_0, \log A_0\} > 0. \quad (13)$$

[proved here]

Proof. At a finite prime p ,

$$\begin{aligned} v_p(u_j) &= r_j \mathbf{1}_{p=2} - s_j m_p, \\ v_p(v_j) &= r_j \mathbf{1}_{p=3} - s_j a_p. \end{aligned}$$

The local projective contribution is therefore

$$\max\{0, -v_p(u_j), -v_p(v_j)\} \log p = s_j \max\left\{0, m_p - \frac{r_j}{s_j} \mathbf{1}_{p=2}, a_p - \frac{r_j}{s_j} \mathbf{1}_{p=3}\right\} \log p.$$

Only primes dividing $6MA$ occur. Since $r_j/s_j \rightarrow \beta$, summing the finitely many terms gives $s_j \kappa + o(s_j)$. At the archimedean place, $u_j = 2^{\varepsilon_j}$ and $v_j = 3^{\varepsilon_j}$, so the contribution is $O(|\varepsilon_j|) = o(1)$. This proves (12).

The inequalities $h(1 : u : v) \geq h(u)$ and $h(1 : u : v) \geq h(v)$, together with (9)–(10), imply (13). $\square$

Remark 3.2 (What κ measures). The real magnitudes of u_j and v_j contain essentially no height: both approach one. The coefficient κ is the surviving denominator slope after all common projective content among $1, u_j, v_j$ has been removed. It is therefore already a *primitive* quantity. Raw numerator sizes are larger and can be misleading.

Remark 3.3 (A useful defect identity). Set

$$\lambda = \log M + \log A - \kappa. \quad (14)$$

Then

$$\lambda = \sum_p (m_p + a_p - \max\{0, m_p - \beta \mathbf{1}_{p=2}, a_p - \beta \mathbf{1}_{p=3}\}) \log p. \quad (15)$$

For $p \neq 2, 3$, the local coefficient is simply $\min(m_p, a_p)$. Thus away from the two structural primes, the linear cancellation rate is exactly the logarithm of the common prime part of M and A .

4 S -unit secants and exact primitive height

4.1 Corvaja–Zannier input

We use the following classical consequence of the Subspace Theorem.

Theorem 4.1 (Corvaja–Zannier, 2005). *Let $\Gamma \subset \mathbb{G}_m^2(\mathbb{Q})$ be finitely generated. As multiplicatively independent $(u, v) \in \Gamma$ satisfy $\max\{h(u), h(v)\} \rightarrow \infty$,*

$$h\left(\frac{u-1}{v-1}\right) \sim h(1 : u : v). \quad (16)$$

More generally, for a fixed nonconstant $F \in \mathbb{Q}(X, Y)$, outside a finite union of translates of proper subtori, $h(F(u, v))$ is bounded below by a positive constant times $\max\{h(u), h(v)\}$. [classical input]

The cited paper states (16) as Corollary 1 and the rational-function statement as its Main Theorem. The exceptional translates are equations $X^c Y^d = w$.

Lemma 4.2 (Coset avoidance). *The sequence (u_j, v_j) meets every fixed proper translate of a subtorus of $\mathbb{G}_m^2$ only finitely many times.*

Proof. A one-dimensional translate has an equation $X^c Y^d = w$ with $(c, d) \neq (0, 0)$. If infinitely many (u_j, v_j) lie on it, then

$$(2^c 3^d)^{\varepsilon_j} = w$$

for infinitely many j . Letting $j \rightarrow \infty$ gives $w = 1$. Since $2^c 3^d \neq 1$ and $\varepsilon_j \neq 0$, the left side is never 1, a contradiction. $\square$

4.2 The principal secant theorem

Define

$$\sigma_j = \frac{u_j - 1}{v_j - 1}. \quad (17)$$

The signs of numerator and denominator agree, so $\sigma_j > 0$.

Theorem 4.3 (Exact primitive secant law). *Under the counterexample hypothesis,*

$$h(\sigma_j) = \kappa s_j + o(s_j). \quad (18)$$

If $\sigma_j = P_j/Q_j$ in lowest terms with $P_j, Q_j \in \mathbb{N}$, then

$$\log P_j = \log Q_j = \kappa s_j + o(s_j). \quad (19)$$

Finally,

$$\sigma_j = \frac{\log 2}{\log 3} + \frac{\log 2(\log 2 - \log 3)}{2 \log 3} \varepsilon_j + O(\varepsilon_j^2). \quad (20)$$

[proved here]

Proof. Lemma 2.1, equations (9)–(10), and Theorem 4.1 give

$$h(\sigma_j) \sim h(1 : u_j : v_j).$$

Theorem 3.1 proves (18). Since $\sigma_j \rightarrow \log 2 / \log 3 \in (0, \infty)$, the reduced numerator and denominator differ by only a bounded multiplicative factor. Their maximum has logarithm $h(\sigma_j)$, proving (19).

For (20), write $a = \log 2$, $b = \log 3$ and use

$$\sigma_j = \frac{e^{a\varepsilon_j} - 1}{e^{b\varepsilon_j} - 1}.$$

Division of the two Taylor series gives

$$\frac{e^{at} - 1}{e^{bt} - 1} = \frac{a}{b} + \frac{a(a-b)}{2b}t + O(t^2).$$

□

The limit $\log 2 / \log 3$ is transcendental: if it were algebraic irrational, Gelfond–Schneider applied to $3^{\log 2 / \log 3} = 2$ would give a contradiction; if it were rational, unique factorization would give a nontrivial equality between powers of 2 and 3.

Corollary 4.4 (Height exponent zero). *There are constants $c, C > 0$ such that*

$$s_j^{-C} \ll \left| \sigma_j - \frac{\log 2}{\log 3} \right| \ll s_j^{-1}, \quad (21)$$

and

$$\frac{-\log \left| \sigma_j - \frac{\log 2}{\log 3} \right|}{h(\sigma_j)} \rightarrow 0. \quad (22)$$

[proved here]

Proof. The coefficient of ε_j in (20) is nonzero. Apply (3), (4), and (18). □

The secants do not merely have “large denominators.” Their *reduced* numerator and denominator have an exact exponential rate κ . Continued fractions improve the real error only polynomially. Thus no irrationality-measure argument can be won by taking these secants at face value: the approximation exponent measured in their true height is zero.

4.3 Directional secants

The principal secant is one member of a toric family. Let $\mathbf{c} = (c_1, c_2)$ and $\mathbf{d} = (d_1, d_2)$ be two linearly independent vectors in $\mathbb{Z}^2$, and put

$$U_j = u_j^{c_1} v_j^{c_2}, \quad V_j = u_j^{d_1} v_j^{d_2}.$$

Then

$$U_j = (2^{c_1} 3^{c_2})^{\varepsilon_j}, \quad V_j = (2^{d_1} 3^{d_2})^{\varepsilon_j},$$

and U_j, V_j are multiplicatively independent. Corvaja–Zannier therefore gives

$$h\left(\frac{U_j - 1}{V_j - 1}\right) \sim h(1 : U_j : V_j) = \kappa_{\mathbf{c}, \mathbf{d}} s_j + o(s_j), \quad (23)$$

where

$$\kappa_{\mathbf{c}, \mathbf{d}} = \sum_p \max\{0, L_{\mathbf{c}}(p), L_{\mathbf{d}}(p)\} \log p, \quad (24)$$

with

$$L_{\mathbf{c}}(p) = c_1 m_p + c_2 a_p - \beta(c_1 \mathbf{1}_{p=2} + c_2 \mathbf{1}_{p=3})$$

and similarly for $\mathbf{d}$. The secant tends to

$$\frac{\log(2^{c_1} 3^{c_2})}{\log(2^{d_1} 3^{d_2})}$$

whenever the denominator is nonzero, again with only polynomial error and linear primitive height. Thus changing toric directions does not evade the barrier.

5 Cross-residual gcds: exact total rate and auxiliary-prime extinction

5.1 Raw residuals

Put

$$E_{2,j} = 2^{r_j} - M^{s_j}, \quad E_{3,j} = 3^{r_j} - A^{s_j}. \quad (25)$$

Both are nonzero and have the sign of ε_j . Define the positive cross-multiplied integers

$$X_j = A^{s_j} |E_{2,j}|, \quad Y_j = M^{s_j} |E_{3,j}|. \quad (26)$$

Then $\sigma_j = X_j/Y_j$.

Lemma 5.1 (Raw size). *One has*

$$\log X_j = \log Y_j = (\log M + \log A)s_j + O(\log s_j). \quad (27)$$

Proof. Since $2^{r_j} = M^{s_j} e^{\varepsilon_j \log 2}$,

$$|E_{2,j}| = M^{s_j} |e^{\varepsilon_j \log 2} - 1|.$$

Equations (3) and (4) imply $\log |e^{\varepsilon_j \log 2} - 1| = O(\log s_j)$. Multiplication by A^{s_j} gives the first formula; the second is identical. $\square$

5.2 The exact total gcd law

Theorem 5.2 (Primitive cancellation identity). *Let*

$$G_j = \gcd(X_j, Y_j). \quad (28)$$

Then

$$\boxed{\log G_j = (\log M + \log A - \kappa)s_j + o(s_j)}. \quad (29)$$

[proved here]

Proof. Writing $\sigma_j = P_j/Q_j$ in lowest terms gives $P_j = X_j/G_j$ and $Q_j = Y_j/G_j$. Subtract (19) from (27). $\square$

This theorem quantifies a phenomenon that raw formulas obscure. The numerator and denominator in (17) possess a common divisor of exponential size, but that size is not free arithmetic information: after division, the surviving height still has the positive linear slope κ .

5.3 No exponential common divisor from new primes

Let S be the finite set of rational primes dividing $6MA$. Define the prime-to- S common residual divisor

$$g_j^{(S)} = \prod_{p \notin S} p^{\min\{v_p(E_{2,j}), v_p(E_{3,j})\}}. \quad (30)$$

Theorem 5.3 (Auxiliary-prime extinction). *For every $\delta > 0$ and all sufficiently large j ,*

$$\log g_j^{(S)} < \delta s_j. \quad (31)$$

Equivalently,

$$\log g_j^{(S)} = o(s_j). \quad (32)$$

[proved here]

Proof. At $p \notin S$, the denominators M^{s_j} and A^{s_j} are p -adic units, so

$$v_p(u_j - 1) = v_p(E_{2,j}), \quad v_p(v_j - 1) = v_p(E_{3,j}).$$

The generalized-gcd form of Corvaja–Zannier’s Corollary 1 says that for multiplicatively independent (u_j, v_j) the total common local proximity of $u_j - 1$ and $v_j - 1$ is less than $\delta \max\{h(u_j), h(v_j)\}$ outside a finite set. Restricting the nonnegative gcd sum to $p \notin S$ can only decrease it. Equations (9)–(10) then give (31); since δ is arbitrary, (32) follows. $\square$

Corollary 5.4 (Where the exponential gcd lives). *The prime-to- S part of G_j is $\exp(o(s_j))$. Hence the full linear rate $\log M + \log A - \kappa$ in (29) is carried by the fixed structural primes in S .*

Proof. Outside S , the factors M^{s_j} and A^{s_j} in (26) contribute no valuation, so the prime-to- S part of G_j is exactly $g_j^{(S)}$. $\square$

Remark 5.5 (Finite families). Take finitely many nonproportional toric directions $\mathbf{c}_1, \dots, \mathbf{c}_t \in \mathbb{Z}^2$ and the residuals $u_j^{c_{i,1}} v_j^{c_{i,2}} - 1$. The prime-to- S gcd of all cleared residuals is bounded by the gcd of any multiplicatively independent pair, and is therefore $\exp(o(s_j))$. Thus no fixed toric jet can acquire an exponential auxiliary common divisor.

The total cross-residual gcd is large, but its linear part is completely structural. This rules out a common proposed escape hatch: manufacture many near-identity residuals and hope that fresh primes clear the denominator. Fresh-prime clearing is subexponential on every fixed finite toric configuration.

6 Fixed-order Richardson extrapolation

The first secant has error $O(\varepsilon_j)$. It is natural to cancel successive Taylor terms by combining secants at $\varepsilon_j, 2\varepsilon_j, \dots, m\varepsilon_j$. The next theorem shows exactly why every fixed order still fails.

6.1 The analytic coefficients

Set

$$\phi(t) = \frac{2^t - 1}{3^t - 1} \quad (t \neq 0), \quad \phi(0) = \rho := \frac{\log 2}{\log 3}. \quad (33)$$

This is real analytic at 0. Write

$$\phi(t) = \sum_{k \geq 0} c_k t^k. \quad (34)$$

Lemma 6.1 (No vanished Taylor coefficient). *Every coefficient c_k is nonzero.*

Proof. Put $a = \log 2$ and $b = \log 3$. From

$$\frac{e^{at} - 1}{e^{bt} - 1} = \frac{e^{at} - 1}{bt} \cdot \frac{bt}{e^{bt} - 1}$$

and the Bernoulli expansion $z/(e^z - 1) = \sum_{n \geq 0} B_n z^n / n!$, the coefficient c_k is a homogeneous rational polynomial in a, b divided by b . Its term of highest degree in a is

$$\frac{a^{k+1}}{b(k+1)!},$$

so the polynomial is not identically zero. After substituting $b = \theta a$ with $\theta = \log 3 / \log 2$ transcendental, a nonzero rational polynomial in θ cannot vanish. $\square$

6.2 The extrapolants

For fixed $m \geq 1$, define

$$R_m(t) = \sum_{i=1}^m (-1)^{i-1} \binom{m}{i} \phi(it). \quad (35)$$

At $t = \varepsilon_j$ every summand is rational:

$$\phi(i\varepsilon_j) = \frac{u_j^i - 1}{v_j^i - 1} \in \mathbb{Q}. \quad (36)$$

Thus $R_m(\varepsilon_j) \in \mathbb{Q}$.

Theorem 6.2 (Exact fixed-order barrier). *For every fixed $m \geq 1$ there are nonzero constants C_m and positive constants c'_m, C'_m such that, as $j \rightarrow \infty$,*

$$R_m(\varepsilon_j) - \rho = C_m \varepsilon_j^m + O(\varepsilon_j^{m+1}), \quad (37)$$

$$c'_m s_j \leq h(R_m(\varepsilon_j)) \leq C'_m s_j, \quad (38)$$

$$\frac{-\log |R_m(\varepsilon_j) - \rho|}{h(R_m(\varepsilon_j))} \rightarrow 0. \quad (39)$$

[proved here]

Proof. The binomial moment identity gives

$$\begin{aligned} \sum_{i=1}^m (-1)^{i-1} \binom{m}{i} &= 1, \\ \sum_{i=1}^m (-1)^{i-1} \binom{m}{i} i^k &= 0 \quad (1 \leq k < m), \\ \sum_{i=1}^m (-1)^{i-1} \binom{m}{i} i^m &= (-1)^{m-1} m!. \end{aligned}$$

Substitution into (34), together with Lemma 6.1, proves (37) with $C_m = (-1)^{m-1} m! c_m \neq 0$.

Now define the fixed rational function

$$F_m(X, Y) = \sum_{i=1}^m (-1)^{i-1} \binom{m}{i} \frac{X^i - 1}{Y^i - 1} \in \mathbb{Q}(X, Y). \quad (40)$$

It is nonconstant: along $(X, Y) = (2^t, 3^t)$ it tends to ρ as $t \rightarrow 0$ but tends to 0 as $t \rightarrow +\infty$. Corvaja–Zannier’s Main Theorem gives a positive linear lower bound for $h(F_m(u_j, v_j))$ outside finitely many subtorus translates. Lemma 4.2 removes those exceptions. The standard height upper bound for a fixed rational map gives the other inequality in (38). Finally, (4), (3), and (37) show that the numerator in (39) is $O(\log s_j)$, while the denominator is $\Theta(s_j)$. $\square$

Remark 6.3 (What increasing the fixed order accomplishes). The construction genuinely improves contact: order m produces m th-order error. The failure is arithmetic, not analytic. Each fixed F_m still has linear primitive height on the S -unit orbit. Consequently no finite m , however large, changes the height exponent.

Caution 6.4 (Growing order is the surviving loophole). Theorem 6.2 is uniform only after m is fixed. If $m = m_j \rightarrow \infty$, the rational function, its exceptional subtori, its coefficients, and its height constants all change. That is not a technical omission: growing complexity is precisely the only regime not eliminated by the theorem. Any proof along this line must control the *reduced* height of the growing family, not merely a common denominator before cancellation.

7 Reciprocal filters and the half-contact theorem

A second natural amplifier uses products of cyclotomic polynomials or other reciprocal polynomials to cancel many leading powers at infinity. Reciprocity imposes a rigid ceiling that appears not to have been used in Report A.

7.1 The algebraic ceiling

A polynomial $P \in \mathbb{Z}[Z]$ of degree D is *reciprocal* if

$$Z^D P(Z^{-1}) = P(Z). \quad (41)$$

Theorem 7.1 (Reciprocal half-contact). *Let $P, Q \in \mathbb{Z}[Z]$ be distinct monic reciprocal polynomials of the same degree D and with constant term 1. Suppose*

$$\frac{P(Z)}{Q(Z)} = 1 + cZ^{-K} + O(Z^{-K-1}), \quad c \neq 0. \quad (42)$$

Then

$$K \leq \left\lfloor \frac{D}{2} \right\rfloor. \quad (43)$$

Equality can occur only if D is even and

$$P(Z) - Q(Z) = cZ^{D/2}. \quad (44)$$

Conversely, (44) gives equality. [\[proved here\]](#)

Proof. Let $H = P - Q$. Because P and Q are monic with constant term 1, the leading and constant coefficients of H vanish. Reciprocity gives

$$Z^D H(Z^{-1}) = H(Z),$$

so the coefficient h_i of Z^i equals h_{D-i} . If d is the largest index with $h_d \neq 0$, the smallest such index is $D - d$. Hence $D - d \leq d$, or $d \geq D/2$. Since $Q(Z) \sim Z^D$, the contact order in (42) is $K = D - d$, proving $K \leq D/2$. Equality forces the smallest and largest nonzero indices of H both to be $D/2$, so H is a single middle monomial. The converse is immediate. $\square$

7.2 A two-scale integer determinant

For fixed P, Q as above, define

$$\Delta_s(P, Q) = P(N^s)Q(M^s) - Q(N^s)P(M^s) \in \mathbb{Z}, \quad N := A = M^\theta, \quad (45)$$

where $\theta = \log 3 / \log 2$ and $A = M^\theta$ under the counterexample hypothesis.

Proposition 7.2 (Exact growth of a fixed reciprocal filter). *With K as in (42),*

$$\log |\Delta_s(P, Q)| = (D(1 + \theta) - K)s \log M + O(1). \quad (46)$$

In particular, by Theorem 7.1,

$$\log |\Delta_s(P, Q)| \geq D \left(\theta + \frac{1}{2} \right) s \log M + O(1). \quad (47)$$

Thus every fixed reciprocal filter produces a rapidly growing nonzero integer rather than a subunit contradiction. [\[proved here\]](#)

Proof. Write $R = P/Q$. Then

$$\Delta_s = Q(N^s)Q(M^s)(R(N^s) - R(M^s)).$$

Since $Q(Z) = Z^D(1 + O(Z^{-1}))$ and $R(Z) = 1 + cZ^{-K} + O(Z^{-K-1})$,

$$\begin{aligned} Q(N^s)Q(M^s) &= M^{D(1+\theta)s}(1 + o(1)), \\ R(N^s) - R(M^s) &= -cM^{-Ks}(1 + o(1)). \end{aligned}$$

Multiplication gives (46); (47) follows from $K \leq D/2$. $\square$

Remark 7.3. The proposition addresses a fixed filter evaluated deeper and deeper on the conditional orbit. It does not rule out a sequence P_s, Q_s of increasing degree with output-dependent coefficient cancellation. Once again, the only surviving regime is growing complexity.

8 Cyclotomic specialization

For $n > 1$, the cyclotomic polynomial Φ_n is monic, reciprocal, and has constant term 1. If $\varphi(n) = \varphi(m) = D$, Theorem 7.1 applies to $P = \Phi_n, Q = \Phi_m$.

8.1 The Möbius-layer description

The standard factorization gives, as a Laurent expansion at infinity,

$$\Phi_n(Z) = Z^{\varphi(n)} \prod_{d|n} (1 - Z^{-d})^{\mu(n/d)}. \quad (48)$$

For equal totients define

$$c_d(n, m) = \mathbf{1}_{d|n} \mu(n/d) - \mathbf{1}_{d|m} \mu(m/d). \quad (49)$$

Then

$$\frac{\Phi_n(Z)}{\Phi_m(Z)} = \prod_{d \geq 1} (1 - Z^{-d})^{c_d(n, m)}. \quad (50)$$

If

$$K(n, m) = \min\{d \geq 1 : c_d(n, m) \neq 0\}, \quad (51)$$

then the first unequal Möbius layer is exactly the contact order at infinity.

Corollary 8.1 (Cyclotomic half-contact). *For distinct $n, m > 1$ with $\varphi(n) = \varphi(m) = D$,*

$$K(n, m) \leq \left\lfloor \frac{D}{2} \right\rfloor. \quad (52)$$

Equality forces $\Phi_n - \Phi_m$ to be a single middle monomial. [\[proved here\]](#)

8.2 Sharp infinite families

Two elementary families attain the ceiling.

For $a \geq 2$, let $h = 2^{a-2}$. Then

$$\Phi_{2^a}(Z) = Z^{2h} + 1, \quad (53)$$

$$\Phi_{3 \cdot 2^{a-1}}(Z) = Z^{2h} - Z^h + 1, \quad (54)$$

so their difference is Z^h and $K = D/2 = h$.

For $a \geq 1$, let $h = 3^{a-1}$. Then

$$\Phi_{3^a}(Z) = Z^{2h} + Z^h + 1, \quad (55)$$

$$\Phi_{2 \cdot 3^a}(Z) = Z^{2h} - Z^h + 1, \quad (56)$$

so their difference is $2Z^h$ and again $K = D/2 = h$. The pair $(3, 4)$ is another sharp example:

$$\Phi_3(Z) - \Phi_4(Z) = Z.$$

8.3 Finite scan through 50000

Computation 8.2 (Equal-totient cyclotomic scan). The script in Appendix D enumerates all pairs

$$3 \leq n < m \leq 50000, \quad \varphi(n) = \varphi(m),$$

computes $K(n, m)$ from (49), and checks the ratio $K/\varphi(n)$. It produced:

Quantity	Value
Equal-totient pairs tested	665095
Maximum $K/\varphi(n)$	$1/2$
Pairs attaining $1/2$	24
Pairs attaining the next ratio $1/4$	34

The 24 sharp pairs are exactly the members, within the search range, of the two families above together with (3, 4). This is finite evidence only; no infinite classification is asserted. [\[computation\]](#)

The sharp pairs are listed in Appendix C. The finite result is useful as a sanity check on the algebraic theorem and as evidence that the middle-monomial equality case is highly rigid.

Reciprocity turns a potentially large coefficient space into a one-dimensional obstruction: two equal-degree reciprocal filters cannot erase more than half the degree at infinity. In the conditional two-scale determinant, even this optimal contact leaves an exponential integer of rate at least $D(\theta + 1/2)s \log M$.

9 Synthesis: what the new route proves and what it rules out

9.1 The height/contact ledger

The constructions above can be summarized by two quantities. For a rational approximation $z_j \in \mathbb{Q}$ to a fixed real number ξ , define

$$\mathcal{A}_j(\xi) = \frac{-\log |z_j - \xi|}{h(z_j)}. \quad (57)$$

A successful irrationality-measure contradiction requires a positive ratio exceeding the known effective threshold for ξ . The new secant families satisfy

$$\mathcal{A}_j\left(\frac{\log 2}{\log 3}\right) \longrightarrow 0.$$

For fixed Richardson order m , the numerator grows from $O(\log s_j)$ to $mO(\log s_j)$, but the denominator remains $\Theta(s_j)$. No fixed m changes the limit.

The common-residual analysis explains why denominator reduction does not secretly rescue the construction. The reduced height is already $\kappa s_j + o(s_j)$, and every fresh-prime gcd is $\exp(o(s_j))$. The only exponential cancellation occurs at the fixed structural primes.

9.2 Families now excluded

The following strategies cannot close the conjecture without a genuinely new ingredient.

1. A single normalized secant, in any fixed pair of toric directions.
2. Any fixed finite Richardson, Prouhet, or finite-difference combination of toric secants.
3. Any attempt to obtain exponential denominator clearing from primes outside the fixed support of $6MA$ for a fixed finite residual family.
4. Any fixed equal-degree reciprocal filter, including fixed products or quotients of cyclotomic polynomials, evaluated along the powers (M^s, A^s) .
5. Any argument based on the raw numerator or common denominator before primitive reduction. The exact slope κ is the relevant invariant.

9.3 Why this is still not a proof

Corvaja–Zannier is a lower-height theorem outside algebraic subtori. Our orbit avoids those subtori, so fixed rational functions retain height. But the theorem is not uniform over a sequence of rational functions whose degrees, supports, and coefficients grow with j . Such a sequence may have output-specific cancellations invisible to every fixed-function statement.

Likewise, auxiliary-prime gcds are subexponential, but structural-prime valuations can be linear and are not bounded away by the Subspace Theorem. A proof may still exist in a carefully designed 2- and 3-adic construction, a growing Hermite–Padé family, or a hybrid in which the same coefficients create both high archimedean contact and primitive structural-prime cancellation.

10 The surviving research target

10.1 A primitive height/contact amplifier

The secant law suggests a precise target.

Research target 10.1 (Growing rational observable). Construct rational functions $F_j \in \mathbb{Q}(X, Y)$ and convergents $r_j/s_j \rightarrow \beta$ such that, for $z_j = (u_j, v_j)$,

1. $F_j(z_j) \in \mathbb{Q}$ tends to a fixed logarithmic constant, preferably $\rho = \log 2 / \log 3$;
2. the analytic contact order k_j along $(2^t, 3^t)$ tends to infinity;
3. after complete numerator/denominator reduction,

$$h(F_j(z_j)) = o(k_j \log s_j),$$

or at least is small enough that the resulting approximation beats an explicit irrationality measure for ρ ;

4. the height saving is supported at the structural primes and is not merely a raw common denominator or an auxiliary-prime gcd.

[open target]

Every fixed F fails condition 3 by the Corvaja–Zannier theorem. Every fixed reciprocal cyclotomic filter fails already at the contact stage. Thus a surviving family must be both growing and nontrivially primitive.

10.2 A finite arithmetic criterion

One can formulate a proof-producing version. Let $F = P/Q$ with coprime integer polynomials, and evaluate at a concrete rational point (u, v) from (5). Compute the exact reduced rational $p/q = F(u, v)$ and a rigorous interval enclosure for $|p/q - \rho|$. Any certificate satisfying

$$0 < \left| \frac{p}{q} - \rho \right| < q^{-\mu_*} \quad (58)$$

with μ_* larger than a certified irrationality exponent for ρ would contradict the counterexample hypothesis. Theorems 4.3 and 6.2 show that no fixed support can produce such certificates asymptotically. A search should therefore optimize reduced height directly over growing supports.

10.3 Suggested computational search

A practical experiment can be organized as follows.

1. Choose a symbolic support $\mathcal{S}_j \subset \mathbb{Z}^2$ whose size grows moderately.
2. Impose k_j linear Taylor-contact equations at $(1, 1)$ along $t \mapsto (2^t, 3^t)$.
3. Solve over $\mathbb{Z}$ or over a lattice adapted to the known 2- and 3-adic valuations.
4. Evaluate the resulting numerator and denominator at exact convergent points (u_j, v_j) .
5. Reduce the rational number completely and record its Smith/valuation profile, not its unreduced denominator.
6. Reject every family whose primitive height divided by $k_j \log s_j$ does not decrease.

The report’s theorems provide calibration tests: a correct implementation must reproduce the slope κ for the basic secant, linear height for every fixed Richardson order, and the half-degree ceiling for reciprocal filters.

11 Formalization blueprint

The new elementary and height bookkeeping can be separated cleanly from the classical deep inputs.

11.1 Kernel-suitable finite core

The following statements are suitable for direct formalization in Lean or another proof assistant.

1. Definition of $u(r, s) = 2^r/M^s$ and $v(r, s) = 3^r/A^s$ as rational numbers; proof of multiplicative independence when $r - s\beta \neq 0$.
2. Exact p -adic valuation formulas and the finite-sum formula for $h(1 : u : v)$, with convergence to (11) under $r/s \rightarrow \beta$.
3. Taylor expansion (20) and the finite-difference moment identities used in Theorem 6.2.
4. The Bernoulli coefficient formula and nonvanishing reduction to transcendence of $\log 3/\log 2$.
5. The reciprocal half-contact theorem, including the equality characterization.
6. The Möbius-layer formula (50) and the deduction $K(n, m) \leq \varphi(n)/2$.
7. A certificate checker for the finite scan: totients, Möbius values, divisor lists, and first unequal layer can all be verified by exact integer computation.

11.2 Explicit external interfaces

Two classical results should remain named hypotheses until formal libraries are available.

Two-logarithm lower bound. For nonzero $r \log 2 - s \log M$, a polynomial lower bound in $\max\{|r|, |s|\}$.

Corvaja–Zannier. The secant-height asymptotic and the fixed rational-function lower height outside finitely many proper torus translates.

All deductions from those interfaces are elementary height, valuation, and Taylor algebra. In particular, the exact gcd slope (29) should be formalized as a short corollary once the secant-height interface is assumed.

12 Conclusion

The new route begins with a hypothetical counterexample and follows its continued-fraction convergents in the opposite direction from the usual determinant method. Instead of seeking many integer points at large scale, it compresses two adjacent multiplicative descriptions into a rational S -unit point converging to $(1, 1)$. This makes the archimedean contact excellent and the nonarchimedean height visible.

The resulting picture is exact. The basic rational secant has primitive height $\kappa s + o(s)$; its reduced numerator and denominator each have that rate. The raw cross-residual gcd has a computable complementary rate, and every part arising from new primes is subexponential. Fixed-order extrapolation improves only the power of a polynomially small error. Reciprocal cyclotomic filters lose at least half their nominal contact before arithmetic is even considered.

This does not cross the four-exponentials frontier. It does narrow the battlefield. A successful proof along the new seam must be a growing-complexity, primitive-height construction whose cancellation is tied specifically to the structural primes and to the conditional output pair. Fixed rational observables, fresh-prime gcds, and reciprocal cyclotomic filters are no longer plausible places for the missing exponential gain to hide.

A Detailed height calculation

For completeness, this appendix writes the projective height in elementary integer terms. Let

$$u = \frac{U_1}{V_1}, \quad v = \frac{U_2}{V_2}$$

be reduced positive rationals and let $D = \text{lcm}(V_1, V_2)$. Then

$$h(1 : u : v) = \log \frac{\max\{D, U_1 D/V_1, U_2 D/V_2\}}{\gcd(D, U_1 D/V_1, U_2 D/V_2)}.$$

Applied to (5), one may take, for large j ,

$$\begin{aligned} U_1 &= 2^{r_j - ms_j}, & V_1 &= M_0^{s_j}, \\ U_2 &= 3^{r_j - as_j}, & V_2 &= A_0^{s_j}. \end{aligned}$$

The three unreduced projective coordinates are all archimedean-comparable because $u_j, v_j \rightarrow 1$. Their common content is nevertheless exponential and is exactly what the local maximum in (11) removes. This is why replacing $h(1 : u : v)$ by $\log \text{lcm}(V_1, V_2)$ can overstate the useful height.

At each prime p , the primitive denominator exponent is

$$\max\{0, -v_p(u_j), -v_p(v_j)\},$$

which after division by s_j converges to the local summand in (11). No limiting interchange issue occurs because only finitely many primes divide $6MA$.

B Bernoulli coefficient formula

Let $a, b \neq 0$ and

$$\frac{e^{at} - 1}{e^{bt} - 1} = \sum_{k \geq 0} c_k(a, b) t^k.$$

Using $z/(e^z - 1) = \sum_{n \geq 0} B_n z^n / n!$ gives

$$c_k(a, b) = \frac{1}{b} \sum_{n=1}^{k+1} \frac{a^n}{n!} \frac{B_{k+1-n} b^{k+1-n}}{(k+1-n)!}. \quad (59)$$

The $n = k + 1$ term is $a^{k+1}/(b(k+1)!)$, so the homogeneous polynomial is nonzero. Setting $b = \theta a$ gives

$$c_k(a, \theta a) = a^k R_k(\theta)$$

for a nonzero rational function R_k whose numerator is a nonzero polynomial in θ . Transcendence of θ proves $c_k \neq 0$.

The finite-difference moments follow from

$$\sum_{i=0}^m (-1)^i \binom{m}{i} e^{it} = (1 - e^t)^m.$$

Differentiating k times at $t = 0$ yields zero for $k < m$ and $(-1)^m m!$ for $k = m$.

C Sharp cyclotomic pairs found in the scan

The 24 pairs (n, m) with $3 \leq n < m \leq 50000$, equal totient, and $K(n, m)/\varphi(n) = 1/2$ were

n	m	$\varphi(n)$	K
3	4	2	1
3	6	2	1
4	6	2	1
8	12	4	2
9	18	6	3
16	24	8	4
27	54	18	9
32	48	16	8
64	96	32	16
81	162	54	27
128	192	64	32
243	486	162	81
256	384	128	64
512	768	256	128
729	1458	486	243
1024	1536	512	256
2048	3072	1024	512
2187	4374	1458	729
4096	6144	2048	1024
6561	13122	4374	2187
8192	12288	4096	2048
16384	24576	8192	4096
19683	39366	13122	6561
32768	49152	16384	8192

The ordering in this table is by increasing maximum index, not by totient. Within the finite range, all but $(3, 4)$ are instances of

$$(2^a, 3 \cdot 2^{a-1}) \quad (a \geq 2), \quad (3^a, 2 \cdot 3^a) \quad (a \geq 1).$$

D Finite scan source

The following pure-Python program uses no external package. It computes Euler totients and Möbius values by sieves, groups indices by totient, and compares the finite coefficient maps in (49).

```
from collections import Counter, defaultdict
from fractions import Fraction

LIMIT = 50_000

# Euler phi sieve.
phi = list(range(LIMIT + 1))
for p in range(2, LIMIT + 1):
    if phi[p] == p:
```

```

        for n in range(p, LIMIT + 1, p):
            phi[n] -= phi[n] // p

# Linear Mobius sieve.
mu = [0] * (LIMIT + 1)
least_prime = [0] * (LIMIT + 1)
primes = []
mu[1] = 1
for n in range(2, LIMIT + 1):
    if least_prime[n] == 0:
        least_prime[n] = n
        primes.append(n)
        mu[n] = -1
    for p in primes:
        if p > least_prime[n] or p * n > LIMIT:
            break
        least_prime[p * n] = p
        mu[p * n] = 0 if n % p == 0 else -mu[n]

def divisors(n: int) -> list[int]:
    small, large = [], []
    d = 1
    while d * d <= n:
        if n % d == 0:
            small.append(d)
            if d * d != n:
                large.append(n // d)
        d += 1
    return small + large[::-1]

def layer_map(n: int) -> dict[int, int]:
    return {d: mu[n // d] for d in divisors(n) if mu[n // d] != 0}

groups = defaultdict(list)
for n in range(3, LIMIT + 1):
    groups[phi[n]].append(n)

pair_count = 0
sharp = []
ratio_counts = Counter()

for degree, indices in groups.items():
    if len(indices) < 2:
        continue
    maps = {n: layer_map(n) for n in indices}
    for i, n in enumerate(indices):
        left = maps[n]
        for m in indices[i + 1:]:
            right = maps[m]
            pair_count += 1
            keys = set(left) | set(right)
            contact = min(
                d for d in keys if left.get(d, 0) != right.get(d, 0)
            )
            ratio = Fraction(contact, degree)
            ratio_counts[ratio] += 1
            if 2 * contact == degree:
                sharp.append((n, m, degree, contact))
            if 2 * contact > degree:
                raise AssertionError((n, m, degree, contact))

print("pairs", pair_count)
print("maximum ratio", max(ratio_counts))
print("sharp count", len(sharp))
print("next ratios", sorted(ratio_counts.items(), reverse=True)[:6])
for row in sharp:
    print(*row)

```

The run used CPython 3 and completed in a few seconds on the report-generation host. Its headline output was

```
pairs 665095
maximum ratio 1/2
sharp count 24
next ratios [(1/2, 24), (1/4, 34), (1/6, 22),
              (1/8, 145), (1/10, 25), (1/12, 186)]
```

E Classical inputs and bibliography

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